Author: Getting back to the ways of describing sequences, I would like to point out that the very method chosen to describe a sequence is not of principal importance. One sequence may be described, for the sake of convenience, by a formula for the nth term, and another (as, for example, the Fibonacci sequence), by the recurrence method. What is important, however, is the method used to describe the law of correspondence, i.e. the law by which any natural no. is placed in correspondence with a certain term of the sequence.

- In a number of cases such a law can be formulated only by words. The examples of such cases are shown below:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, \ldots\ldots \quad (12)$$
$$3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots\ldots \quad (13)$$

- In both cases we cannot indicate either the formula for the nth term or the recurrence relation. Nevertheless, you can without great difficulties identify specific laws of correspondence and put them in words.

Sam: Sequence (12) lists all the prime numbers in ascending order. (Matches with Reader) ✓
Sequence (13) lists out the approximations of π in increasing greater degree of approximation. (Matches with Reader) ✓

Reader: It may seem that a numerical sequence differs from a random set of numbers by a presence of an intrinsic degree of order that is reflected either by the formula for the nth term or by the recurrence relation. However, the last two examples show that such a degree of order needn't be present.

Author: Actually, a degree of order determined by a formula (an analytical expression) is not mandatory. It is important, however, to have a law (a rule, a characteristic) of correspondence, which enables one to relate any natural number to a certain term of a sequence. In examples (12) and (13) such laws of correspondence are obvious. ∴ (12) and (13) are not inferior/superior to sequences (1)–(11) which permit an analytical description.

Later we shall talk about the geometric image (or map) of a numerical sequence. Let us take two coordinate axes, x and y. We shall mark on the first axis integers $1, 2, 3, \ldots\ldots, n, \ldots\ldots$ and on the second axis, the corresponding terms of a sequence, i.e. the numbers $y_1, y_2, y_3, \ldots\ldots, y_n, \ldots\ldots$ Then the sequence can be represented by a set of points $M(n, y_n)$ on the coordinate plane.